

# Factory Analytics

A comprehensive technical guide to the architecture, core user workflows, and state management of the Soluperts Flutter application.



# Dual-Role Architecture

## Line Incharge (LI)

Designed for focused operational control. The LI manages a single, assigned assembly line.

Key responsibilities include configuring stations, logging real-time production updates, capturing QC photos, and overseeing inbound/outbound inventory for their specific line.

## Production Incharge (PI)

Designed for broad oversight and analytics. The PI has visibility over all factory assembly lines.

Key responsibilities include verifying line setups, managing factory-wide workforce, conducting global QC reviews, and analyzing production trends through visual charts.



# App Entry & Authentication

The app utilizes a secure, state-driven authentication process integrated deeply with GetX controllers.

- **Dependency Injection:** `NetworkController` and `AuthController` initialize on startup via `GetMaterialApp`.
- **Storage:** Combines SharedPreferences (user types) and Flutter Secure Storage (JWT access/refresh tokens).
- **Smart Routing:** Upon checking the stored auth status, the app instantly routes users to either `HomeProductionIncharge` or `HomePageLineIncharge`.
- **Validation:** Enforces strong password criteria (special char, number, uppercase, 8+ chars).





# Seamless Token Refresh

Maintaining a secure but uninterrupted user experience on the factory floor is critical.

The **ApiServices** class intercepts all HTTP requests. If a backend API returns a **401 Unauthorized** error, the app automatically triggers `refreshToken()`.

It utilizes the securely stored refresh token. Upon success, the new access token is saved and the original request is retried invisibly to the user. If it fails, they are cleanly redirected to the LoginPage.





# Line Incharge Workflows



## Station Setup

Operated via `AssemblyLineModification`. Fetches current assembly line ID and allows the LI to view and modify specific workstation configurations.



## Production Update

Fetches active orders and provides forms to log exact production output, QC pass/fails, input conversion, and per-station backlogs.



## Line QC Photos

Uses the `QualityCheckController` to upload camera captures. Features date/station filters and pull-to-refresh list integration.



# Production Incharge **Workflows**



## Line Verification

Loads all lines globally via `AssemblyLineController`. The PI taps a row to view station setups and officially verify the line configuration.



## Global QC Review

The `ViewQCPiController` populates a master table filtering QC photos by Assembly Line, Station, Clicked By, and Date.



## Trend Analytics

Aggregates production data from all lines. `ChoiceChips` filter data, rendering visualizations using `Syncfusion Flutter Charts`.

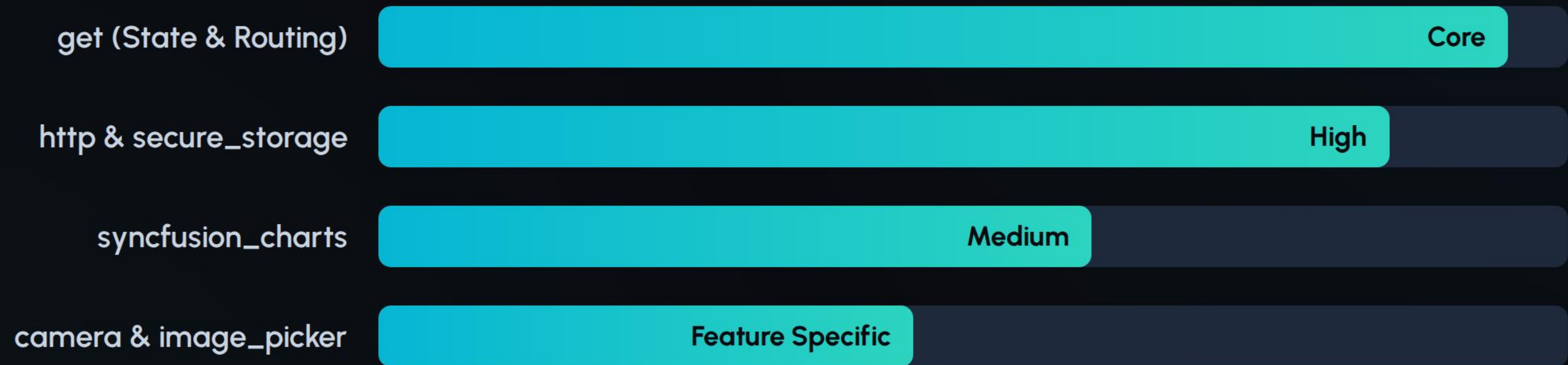


# Shared Components & Utilities

| Component       | File / Class         | Usage Context                                                                                        |
|-----------------|----------------------|------------------------------------------------------------------------------------------------------|
| Employees List  | EmployeesList        | Shared view. LI scopes workers to one line; PI sees global workforce. Allows adding/editing entries. |
| QC Image Viewer | QcImageView          | Shared UI in `view/shared/`. Displays individual quality check photos loaded from APIs.              |
| API Services    | utils/constants.dart | Global HTTP wrapper. Attaches Bearer tokens, manages error states, and handles 401 retries.          |
| Network Monitor | NetworkController    | Listens to `connectivity_plus`. Triggers red persistent snackbar on disconnects.                     |



# Core Package Dependencies



*The app is heavily reliant on GetX for state management and dependency injection, supported by robust networking and hardware access plugins.*



# System Resilience

# 401

## Auto-Recovery Status

## Built for the Factory Floor

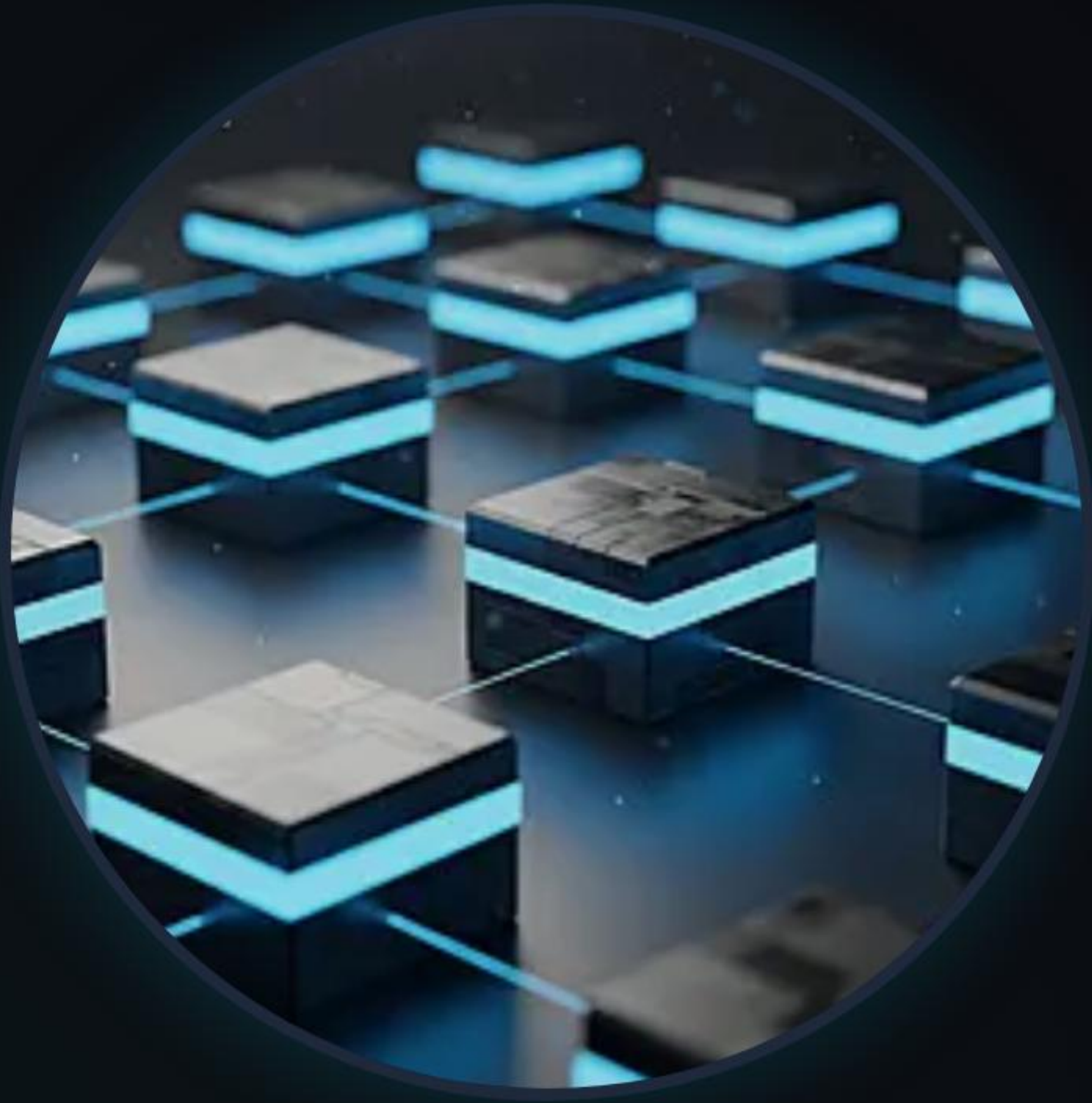
Intermittent connectivity is handled gracefully. The `NetworkController` surfaces immediate UI alerts upon disconnection.

Simultaneously, token expiration (401 response) is caught globally. The app pauses API calls, performs a silent background refresh, and replays the requests.

All critical actions are logged locally via **FLog**, which the Production Incharge can review or export from the `LogsPage` for remote debugging.



# Structured Data Models



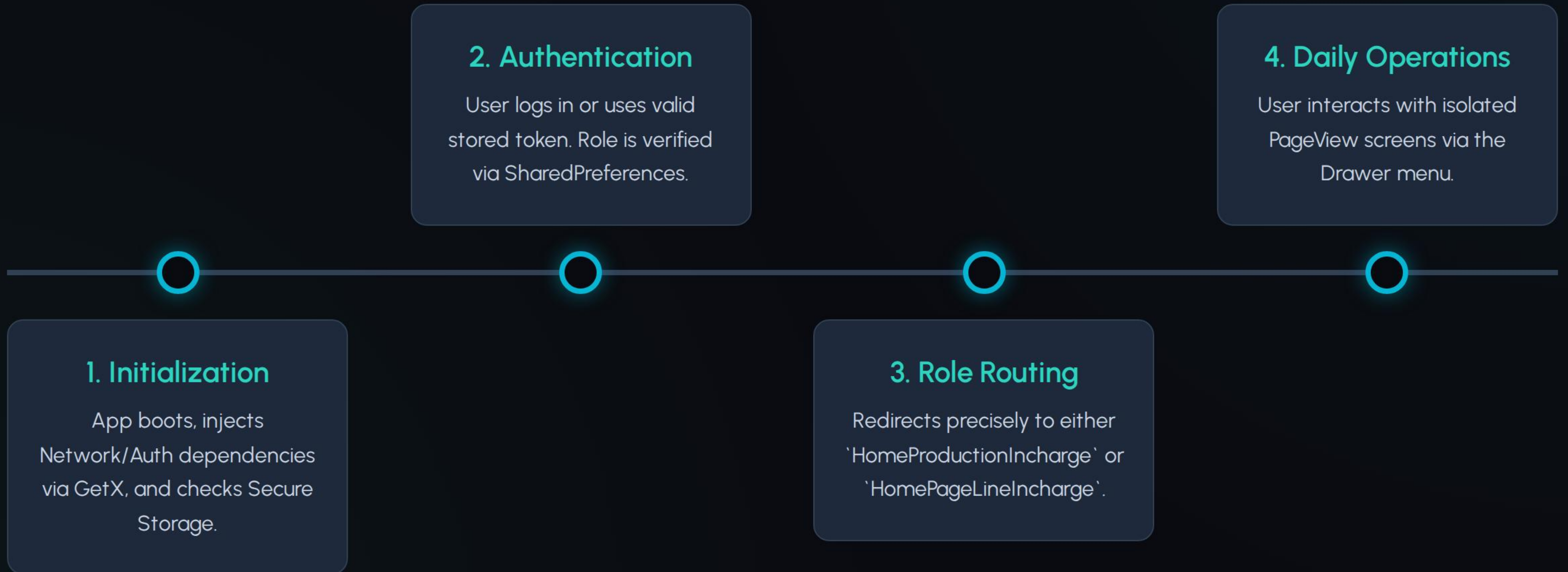
## Strict MVC Architecture

Data is cleanly separated into Models to ensure type safety across the Flutter app:

- > **User & Roles:** UserModel, LineInchargeModel, ProductionInchargeModel handle user identity and scope limits.
- > **Operations:** AssemblyLineModel, StationModel manage the physical factory layout.
- > **Metrics:** ProductionUpdateModel, QcModel, EmployeeModel handle the heavy lifting of daily inputs.



# Standard User Journey





# Questions?

Factory Analytics App Architecture & Implementation

Built with Flutter & GetX State Management



# Image Sources



[https://static.vecteezy.com/system/resources/previews/023/465/537/non\\_2x/locked-padlock-dark-mode-glyph-ui-icon-restrict-access-security-settings-user-interface-design-white-silhouette-symbol-on-black-space-solid-pictogram-for-web-mobile-isolated-illustration-vector.jpg](https://static.vecteezy.com/system/resources/previews/023/465/537/non_2x/locked-padlock-dark-mode-glyph-ui-icon-restrict-access-security-settings-user-interface-design-white-silhouette-symbol-on-black-space-solid-pictogram-for-web-mobile-isolated-illustration-vector.jpg)

Source: [www.vecteezy.com](https://www.vecteezy.com)

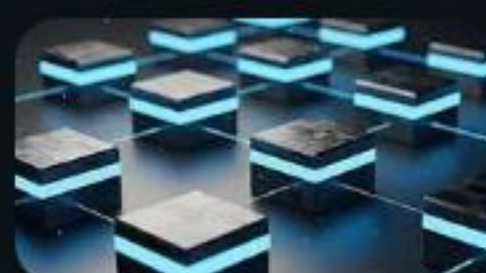
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Source: [pngtree.com](https://pngtree.com)

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